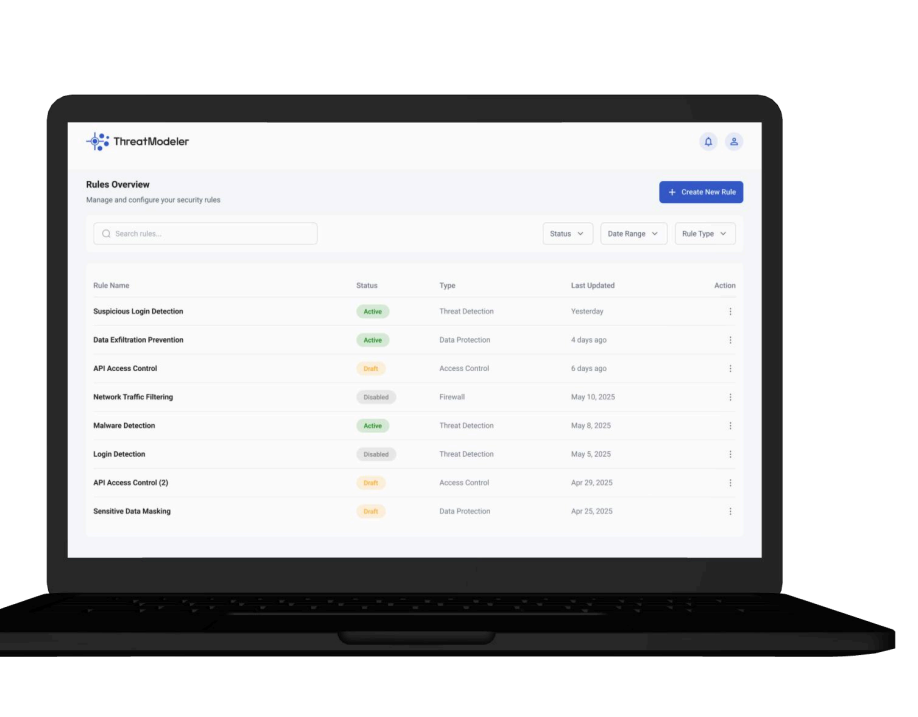
ThreatModeler

UX Case Study



UX Case Study: ThreatModeler: Security Rule Engine

Problem Statement

The current Security Rule Engine in ThreatModeler is overly complex and unintuitive, particularly for users without a strong technical background. This complexity leads to frequent misconfigurations, abandonment of workflows, and a general reluctance to engage with the feature. A lack of clear guidance, structured logic flow, and contextual support creates a frustrating user experience that hinders the effective automation of security policies.

Objectives & Goals

- Understand how users interact with the current rule, building interface and identify points of confusion.
- Conduct contextual research to align the design with real user needs and mental models.
- Design a simplified, guided, and intuitive interface for users of all technical levels.
- Use iterative prototyping and usability testing to refine the experience based on user feedback.

Process

Discover

Define

Ideate

Design

Product Users

Our target audience is the working population of age group 25-45

Business Challenges

- Users find the current interface intimidating or too technical.
- Due to a complex UI, users frequently rely on support teams or request training sessions.
- Users incorrectly set up rules due to lack of clarity or guidance.
- Advanced capabilities (e.g., logic chaining, simulations) are rarely used due to poor discoverability.
- New users take too long to build useful rules or gain confidence.
- A complex or frustrating experience can lead to poor user reviews.

Observation

60%

Users treat triggers and conditions as interchangeable, leading to logic errors.

70%

Users hesitate to create/edit rules, fearing they might cause system issues or security gaps.

45%

Users either ignore or misinterpret feedback messages during rule creation.

65%

Users are unaware of existing rule templates that could save time and reduce errors.

55%

Non-developers think they can't use the tool, even when they can.

User Needs

- Simplified rule creation process for non-technical users
- Better understanding of rule components (Trigger, Condition, Action)
- Quick start options for common use cases
- Visual way to understand and build logic
- Real-time error prevention and guidance
- Access to help without leaving the page

Features & Functionalities

To resolve user needs

Clear Instructions

Easy Navigation

Real-time Status Updates

Product User Challenges

- Guided Rule Creation Wizard with step-by-step flow and plain language
- Inline Tooltips & Descriptions explaining each component in-context
- Pre-built Templates & Example Rules Library
- Drag-and-Drop Rule Builder UI with real-time preview
- Live Validation with Clear Suggestions on incorrect rules
- Undo, Version History, and Save as Draft features
- Commenting, Sharing, and Approval Workflow
- Smart Suggestions and Auto-Complete for triggers/conditions/actions
- Contextual Help Panel or Chat Support Widget

Competitor Analysis

Microsoft Defender for Cloud

Unified cloud security posture management

Feature:

- Uses policy-based rules and recommendations to automate security tasks
- Seamless Azure integration, clear remediation steps, good for semi-technical users

Microsoft Defender

snyk

Snyk

Developer-first security with policy rules

Feature:

- Simplifies security rule configuration for developers with easy integrations and policy enforcement
- Clean UI, excellent onboarding, real-time feedback for errors

Unique Features

- **Smart Rule Assistant (AI-Powered Suggestions):** Suggests possible triggers, conditions, and actions based on user behavior or context.
  Why: Reduces time and mental load, helps non-technical users create correct rules with confidence.
- **Real-Time Impact Simulator:** Before activating a rule, users can preview how it would affect current system workflows or threat models.
  Why: Builds trust, reduces fear of making mistakes, and lowers misconfiguration risk.
- **Visual Logic Map View:** An interactive, drag-and-drop visual layout of rule logic.
  Why: Simplifies complex logic structures into understandable diagrams, ideal for semi-technical users.
- **Natural Language Rule Creator:** Users type rules in plain English, which the system translates into structured logic.
  Why: Removes barriers for non-technical users and makes onboarding much easier.
- **Interactive Onboarding for Rule Engine:** A step-by-step walkthrough that guides users through their first rule with real-time help.
  Why: Ensures activation success, reduces support dependency, and improves first-time experience.

Amit Sharma

Security Analyst

About

30 Gurugram

MBA Employee

Description

Works in a mid-sized fintech company in Gurugram, experienced in threat detection but not in scripting or complex logic

Goals

Automate security rules quickly without writing code or needing engineering support

Challenges

Struggles to understand how to structure rules; unsure if the rules will behave as intended; limited UI guidance

Description

Based in Pune, manages product development lifecycle, limited technical knowledge but responsible for security compliance

Goals

Ensure the product meets security requirements through automation and team collaboration

Challenges

Intimidated by technical jargon, unclear about which rules are needed and how they affect system behavior

Priya Desai

Product Manager

About

34 Pune

MBA Employee

Ravi Verma

DevOps Engineer

About

28 Hyderabad

BTech Employee

Description

Based in Hyderabad, experienced with cloud deployments and CI/CD pipelines

Goals

Integrate security rules into deployment processes to prevent vulnerabilities automatically

Challenges

Finds the UI unintuitive, lacks concrete examples or templates, spends time troubleshooting misconfigured rules

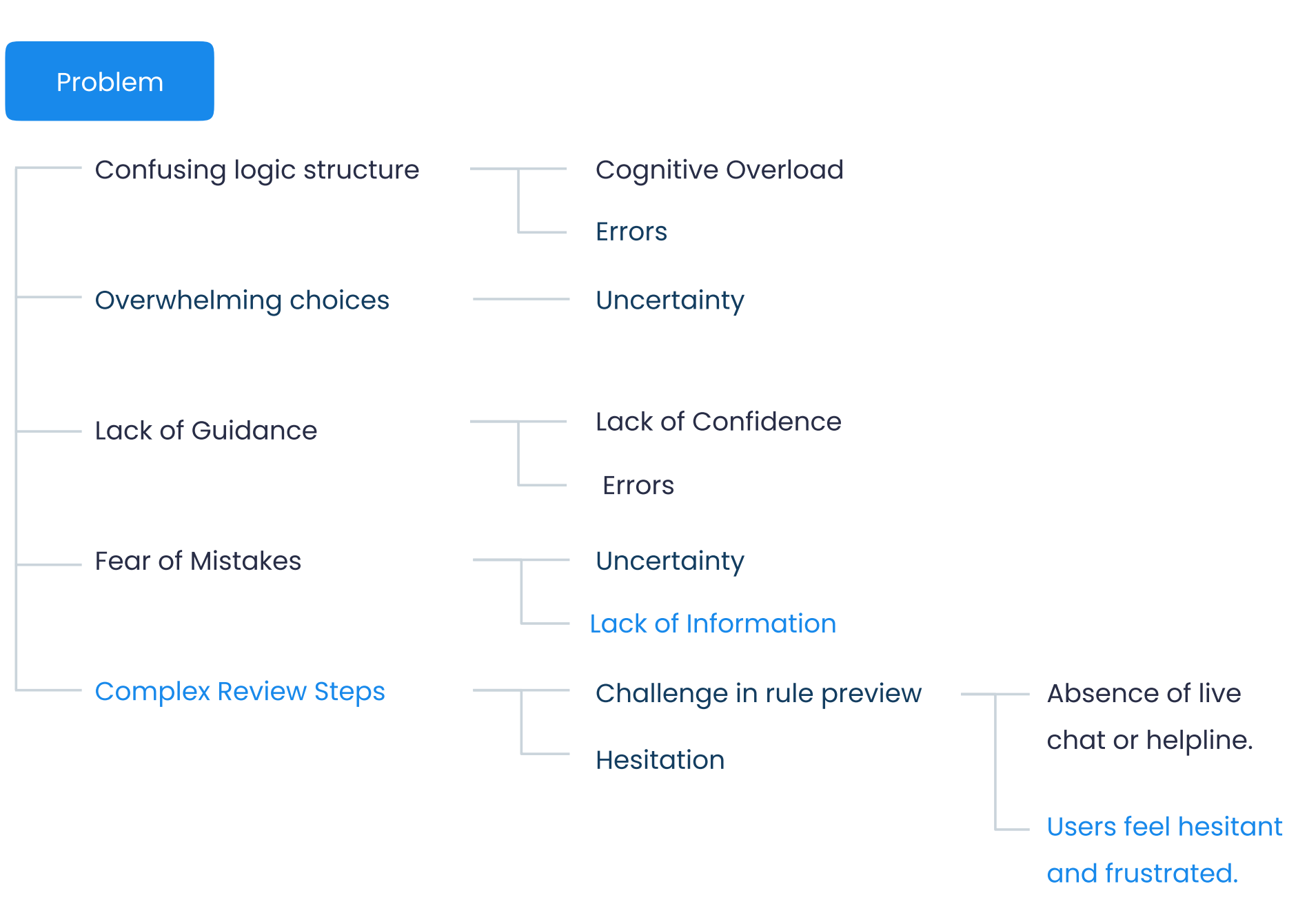
Task Mapping

Step	Step 1	Step 2	Step 3	Step 4
Task	Navigate to Rule Engine	Choose Trigger & Conditions	Define Actions & Logic	Review & Publish Rule
Environment	Dashboard with navigation	Trigger selection interface	Action builder with logic blocks	Summary screen / Publish CTA
Challenges	Unclear entry point	Overwhelming choices	Confusing logic structure	Fear of errors or misconfigurations
Emotions	Hesitation	Frustration	Cognitive overload	Cautious relief
Thoughts	"Where do I begin?"	"Which one do I choose?"	"What if I break something?"	"Is this correct?"
Urgency Level	Medium	High	High	High
Design Opportunity	Improve entry clarity	Guided selection w/ examples	Visual logic templates & validation	ValidationSimulation preview + confidence messaging

Eisen Hover Matrix

	Urgent	Not Urgent
Important	• Simplify trigger/condition logic • Show instant feedback on logic errors • Add guided rule templates	• Introduce tooltips & help docs • Add personalization for rules • Long-term analytics tracking
Not Important	• Color-code sections • Auto-save drafts	• Export rule history as PDF • Allow custom UI themes

Root Cause Analysis (RCA)



Problem

Confusing logic structure — Cognitive Overload, Errors

Overwhelming choices — Uncertainty

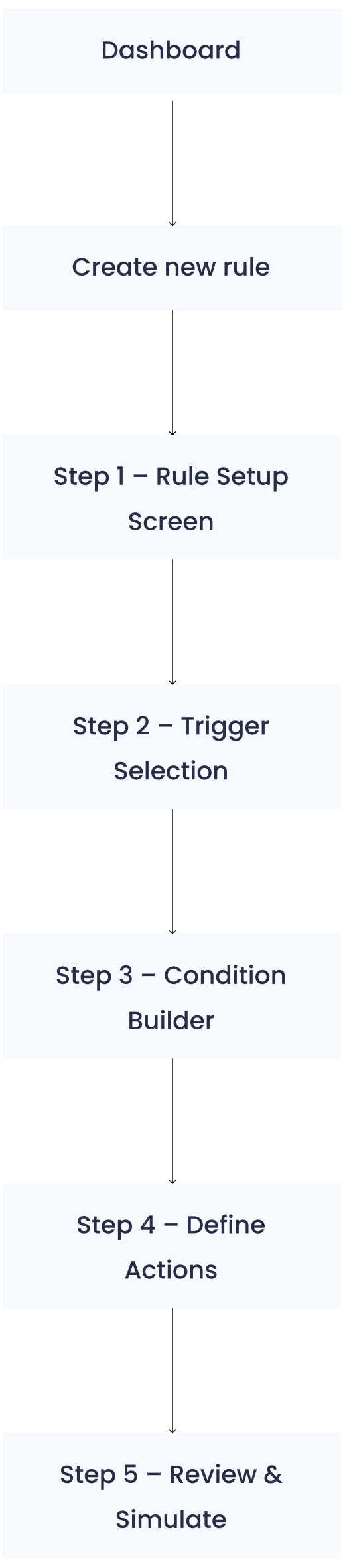
Lack of Guidance — Lack of Confidence, Errors

Fear of Mistakes — Uncertainty, Lack of Information

Complex Review Steps — Challenge in rule preview, Hesitation

Challenge in rule preview — Absence of live chat or helpline., Users feel hesitant and frustrated.

Taskflow



Dashboard

Create new rule

Step 1 – Rule Setup Screen

Step 2 – Trigger Selection

Step 3 – Condition Builder

Step 4 – Define Actions

Step 5 – Review & Simulate

Thank You For Watching